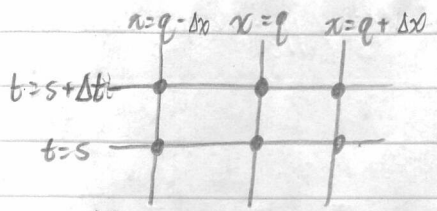
Crank Nicolson Method

- We ultimately want to solve the Time-Dependent Schrödinger Equation:

$$i\hbar \frac{\delta \Psi}{\delta t} = \hat{H}\Psi = -\frac{\hbar^2}{2m}\frac{\delta^2 \Psi}{\delta x^2} + V(x)\Psi$$

- But, to allow our computer to begin to compute the derivatives, we must define a grid:



> At $x=q$ & $t=s$, we can approximate: $\frac{\delta^2\Psi}{\delta x^2} \approx \frac{1}{(\Delta x)^2}\left(\Psi(x=q+\Delta x, t=s) - 2\Psi(x=q, t=s) + \Psi(x=q-\Delta x, t=s)\right)$

> At $x=q$ & $t=s+\Delta t$, we can approximate: $\frac{\delta^2\Psi}{\delta x^2} \approx \frac{1}{(\Delta x)^2}\left(\Psi(x=q+\Delta x, t=s+\Delta t) - 2\Psi(x=q, t=s+\Delta t) + \Psi(x=q-\Delta x, t=s+\Delta t)\right)$

> Since we're taking a small time step, using only the approximation for $\frac{\delta^2\Psi}{\delta x^2}\Big|_{t=s}$ or only the approximation for $\frac{\delta^2\Psi}{\delta x^2}\Big|_{t=s+1}$ won't be particularly accurate. Instead, we will average the two approximate:

$$\frac{\delta^2\Psi}{\delta x^2} \approx \frac{1}{2(\Delta x)^2}\Big(\Psi(q+\Delta x, s) - 2\Psi(q,s) + \Psi(q-\Delta x, s) + \Psi(q+\Delta x, s+\Delta t) - 2\Psi(q, s+\Delta t) + \Psi(q-\Delta x, s+\Delta t)\Big)$$

> Using similar logic, we can approximate Ψ on the right-hand side of the TDSE as:

$$\Psi \approx \frac{1}{2}\left(\Psi(q,s) + \Psi(q, s+\Delta t)\right)$$

> Finally, we can approximate $\frac{\delta\Psi}{\delta t}$ at $x=q$ as:

$$\frac{\delta\Psi}{\delta t} \approx \frac{\Psi(q, s+\Delta t) - \Psi(q,s)}{\Delta t}$$

- We can plug in our approximations to see, at $x=q$:

$$i\hbar\left(\frac{\Psi(q,s+\Delta t) - \Psi(q,s)}{\Delta t}\right) = -\frac{\hbar^2}{2m}\cdot\frac{1}{2(\Delta x)^2}\left(\begin{array}{l}\Psi(q+\Delta x, s) - 2\Psi(q,s) + \Psi(q-\Delta x, s) \\ + \Psi(q+\Delta x, s+\Delta t) - 2\Psi(q, s+\Delta t) + \Psi(q-\Delta x, s+\Delta t)\end{array}\right) + \frac{V(q)}{2}\left(\Psi(q,s) + \Psi(q, s+\Delta t)\right)$$

- Our goal is to have Ψ at $t=s+\Delta t$ on one side of the equation and Ψ at $t=s$ on the other:

$$\Psi(q, s+\Delta t) + \frac{i\Delta t}{\hbar}\left[-\frac{\hbar^2}{2m(\Delta x)^2}\left(\Psi(q+\Delta x, s+\Delta t) - 2\Psi(q, s+\Delta t) + \Psi(q-\Delta x, s+\Delta t)\right) + V(q)\Psi(q, s+\Delta t)\right]$$
$$= \Psi(q,s) - \frac{i\Delta t}{\hbar}\left[-\frac{\hbar^2}{2m(\Delta x)^2}\left(\Psi(q+\Delta x, s) - 2\Psi(q,s) + \Psi(q-\Delta x, s)\right) + V(q)\Psi(q,s)\right]$$

- We can simplify this immensely by writing $\alpha = \frac{i\hbar\Delta t}{4m(\Delta x)^2}$ and $\beta = \frac{i\Delta t}{2\hbar}$, so:

$$-\alpha\Psi(q+\Delta x, s+\Delta t) + \left[1+\beta(t_0+V(q))\right]\Psi(q, s+\Delta t) - \alpha\Psi(q-\Delta x, s+\Delta t)$$
$$= +\alpha\Psi(q+\Delta x, s) + \left[1-\beta(t_0+V(q))\right]\Psi(q,s) + \alpha\Psi(q-\Delta x, s)$$

- Note that at time $t=s$, all of the points in the grid for Ψ are known values. Thus, let $D_1 = (1-\beta(t_0+V(1)))\Psi(1,s) + \alpha\Psi(2,s)$, $D_2 = \alpha\Psi(1,s) + (1-\beta(t_0+V(2)))\Psi(2,s) + \alpha\Psi(3,s)$, ...

$$D_{N-1} = \alpha\Psi(n-2,s) + (1-\beta(t_0+V(n-1)))\Psi(n-1,s)$$

- We can now put together all the math we have done thus far to establish the system of equations:

$$
\begin{cases}
[1+\beta(t_0+V(1))]\Psi(1,s+\Delta t) - \alpha\Psi(2,s+\Delta t) = \heartsuit_1 \\
-\alpha\Psi(1,s+\Delta t) + [1+\beta(t_0+V(2))]\Psi(2,s+\Delta t) - \alpha\Psi(3,s+\Delta t) = \heartsuit_2 \\
\vdots \\
-\alpha\Psi(n-3,s+\Delta t) + [1+\beta(t_0+V(n-2))]\Psi(n-2,s+\Delta t) - \alpha\Psi(n-1,s+\Delta t) = \heartsuit_{n-2} \\
-\alpha\Psi(n-2,s+\Delta t) + [1+\beta(t_0+V(n-1))]\Psi(n-1,s+\Delta t) = \heartsuit_{n-1}
\end{cases}
$$

- As is standard procedure, we can cast the system of equations into a matrix:

$$
\begin{pmatrix}
1+\beta(t_0+V(1)) & -\alpha & 0 & 0 & \cdots & 0 \\
-\alpha & 1+\beta(t_0+V(2)) & -\alpha & 0 & \cdots & 0 \\
0 & -\alpha & 1+\beta(t_0+V(3)) & -\alpha & \ddots & \vdots \\
0 & 0 & -\alpha & \ddots & \ddots & -\alpha \\
\vdots & \vdots & \vdots & \ddots & & \\
0 & 0 & 0 & \cdots & -\alpha & 1+\beta(t_0+V(n-1))
\end{pmatrix}
\begin{pmatrix}
\Psi(0,s+\Delta t) \\ \Psi(1,s+\Delta t) \\ \Psi(2,s+\Delta t) \\ \vdots \\ \Psi(n-1,s+\Delta t)
\end{pmatrix}
=
\begin{pmatrix}
\heartsuit_0 \\ \heartsuit_1 \\ \heartsuit_2 \\ \vdots \\ \heartsuit_{n-1}
\end{pmatrix}
$$

- From here we can just focus on converting the matrix into Reduced Echelon Form, which will take $\mathcal{O}(n)$ time because the matrix is tridiagonal. Afterwards, solving for each Ψ in the grid along $t=s+\Delta t$ becomes simple